

Experiment No. 2

Introduction to Semi-Controlled Rectifiers

Objectives:

1. To analyze resistive and resistive capacitive firing/triggering commercially available silicon-controlled rectifier (SCR).
2. To observe the effect of change in firing angle on output DC voltage.
3. To understand the difference between controllable and uncontrollable devices on output quality.

Equipment:

- AC supply
- Diodes, suitable for your voltage and power levels
- SCR, suitable for your voltage and power levels
- Variable gate resistance (0 – 500k/1M Ω)

Section1: Single Phase Half Wave Semi-Controlled Rectifier using R-Triggering

Arrange the circuit as shown in figure1. Locate a commercially available SCR which can serve load resistor of 10ohms and inductor of 50mH. The circuit is being energized by a 230V(rms), 50 Hz AC source. Mention the part number of the SCR, download its spice model to import in LTspice, and also download its datasheet to conduct the oncoming calculations. Make an educated choice of the diode as well.

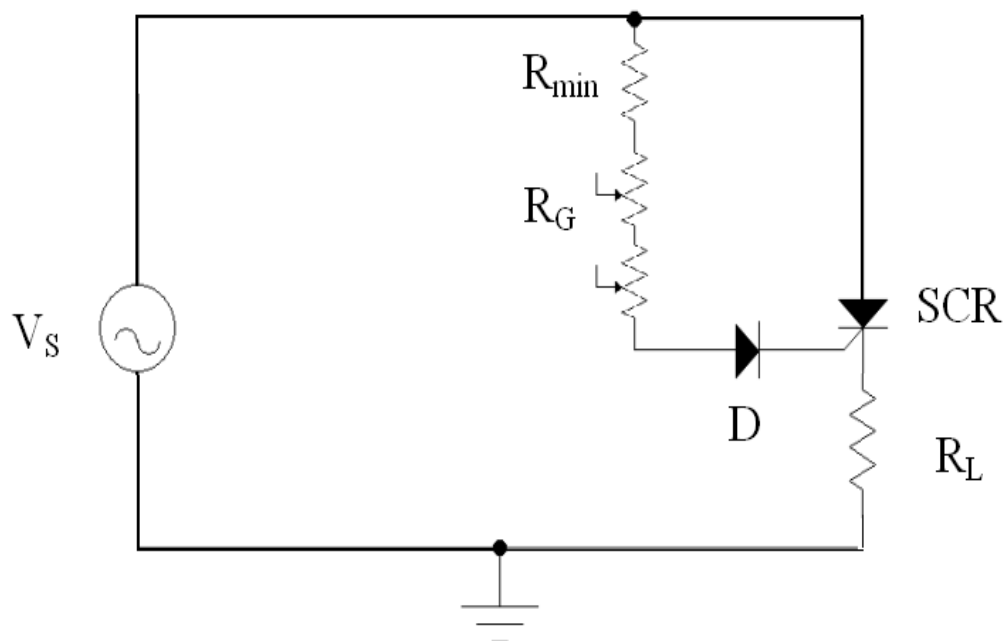


Figure 1

Design Procedure

Minimum Resistance (R_{min}):

The minimum resistance R_{min} should be such that when $R_G = 0$, the gate current does not exceed the maximum allowable value,

$$R_{min} = \frac{V_{s(max)}}{I_{g(max)}} \quad (1)$$

$V_{s(max)} =$ _____ (equal to the maximum voltage supplied by the AC supply)

$I_{g(max)} =$ _____ (gate current is usually given by I_{GT} in the datasheets)

Using Equation (1): $R_{min} =$ _____

The value of the selected standard resistor = $R_{min} =$ _____

Gate Resistance (R_G):

The minimum value of gate triggering current is selected as specified in the datasheet (for instance, it is 80 μ A in case of MCR 100-6). The SCR will turn on when instantaneous value of supply voltage V_s becomes equal to

$$V_s \geq I_{G(min)}[R_G + R_{min}] + V_{G(min)} + V_D \quad (2)$$

$$R_G + R_{min} \leq \frac{V_s - V_{G(min)} - V_D}{I_{G(min)}} \quad (3)$$

$V_D =$ _____ (equal to the forward voltage drop across the diode)

$I_{G(min)} =$ _____ (minimum value of gate current at which SCR fires)

$V_{G(min)} =$ _____ (gate-cathode voltage corresponding to $I_{G(min)}$)

Using Equation (2): $R_G =$ _____

The value of the selected standard resistor = $R_G =$ _____

Selection of SCR:

Maximum value of anode to cathode current from data sheet = _____ A

Latching current from data sheet = _____ A

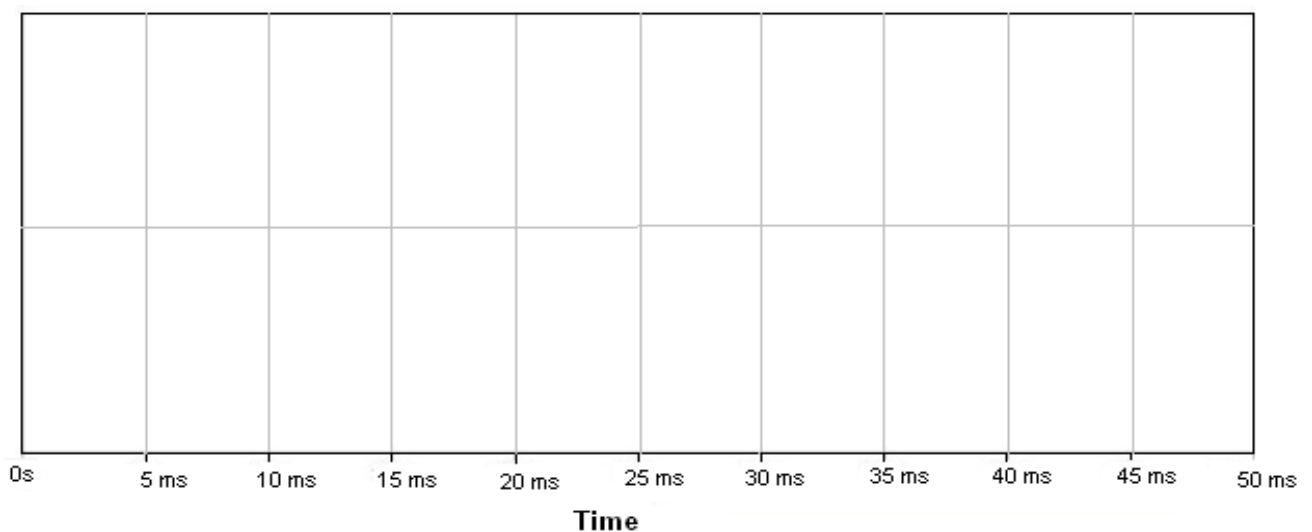
Holding current from data sheet = _____ A

Procedure:

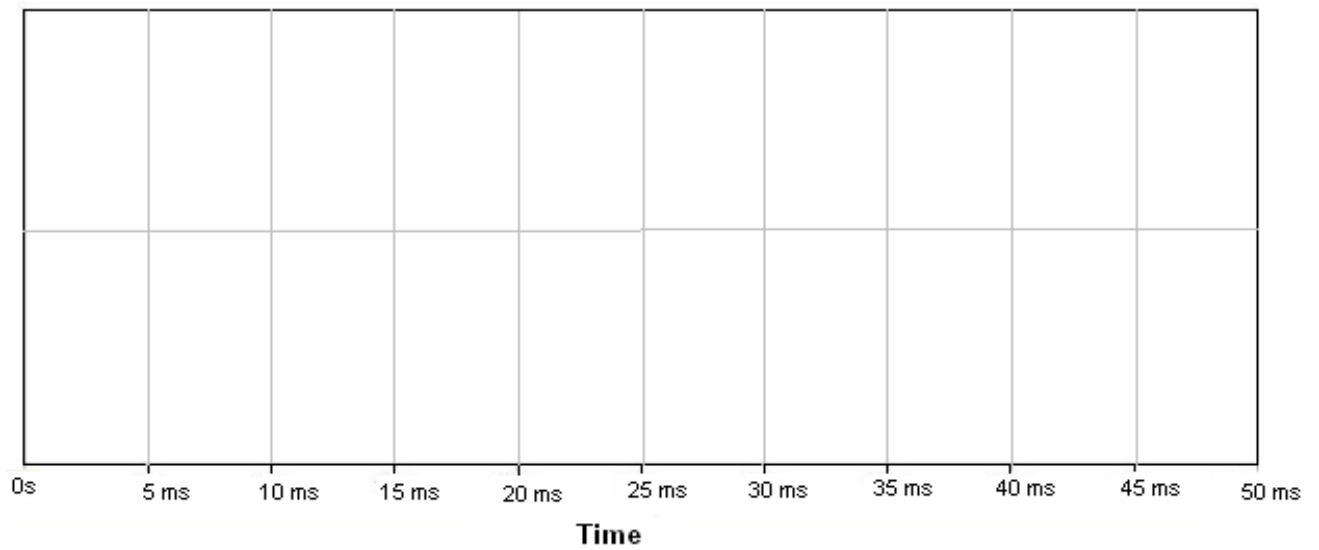
1. Arrange the circuit as shown in Figure 1 and apply the input voltage.
2. Keep the potentiometer resistance R_G to its maximum value so I_G is very small, and SCR is not triggered. Measure and observe the voltage across load and SCR.
3. Decrease the resistance of potentiometer R_G such that SCR is now fired and almost all the input voltage appears across the load. Measure and observe the change in voltage across load and SCR
4. Record your observations for firing angles of 45° and 90° .
5. Record your observations and calculations in tabular form and also plot the waveforms observed on oscilloscope with firing angles **clearly mentioned**.

Observation Table 1:

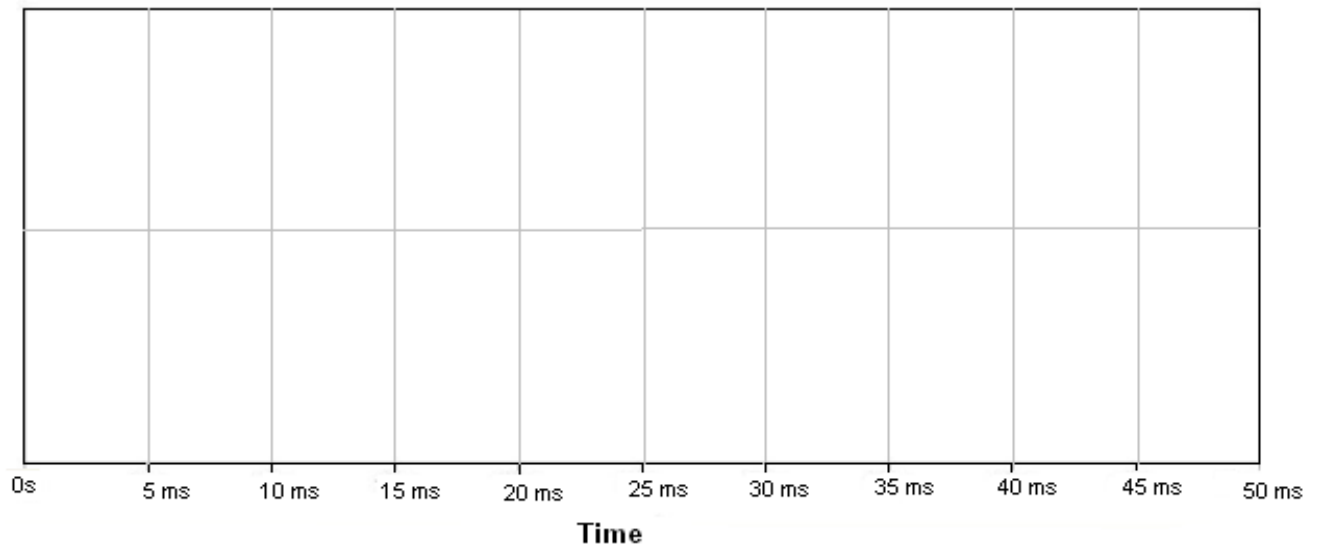
No.	Firing Angle α	V_o (dc) (Measured from Spice Simulations)	V_o (dc) (Theoretically using calculations)
1.			
2.			

1- Sketch voltage waveform across load resistance for 45° firing angle.

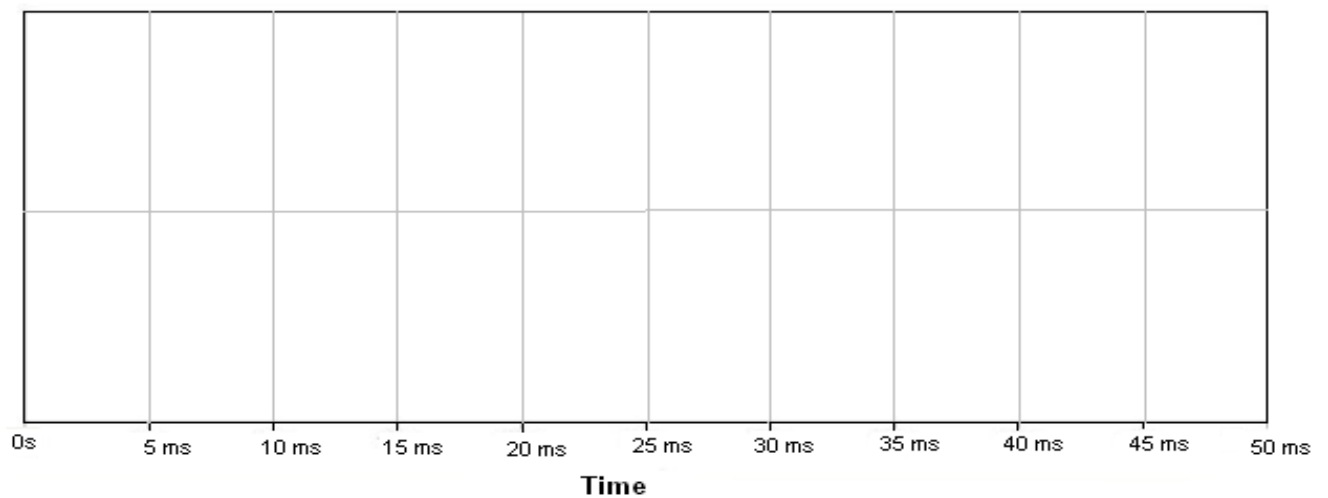
2- Sketch voltage waveform across thyristor (V_{AK}) for 45° firing angle.



3- Sketch voltage waveform across load resistance for 90° firing angle.



4- Sketch voltage waveform across thyristor (V_{AK}) for 90° firing angle.



Show your calculations of output voltage with both the firing angles:

Please identify if there is any need for free-wheeling diode in this application?

Section2: Single-Phase Full Wave Semi-Controlled Rectifier using R-Triggering

Arrange the circuit as shown in figure2. Borrow the procedure and components from section1 above.

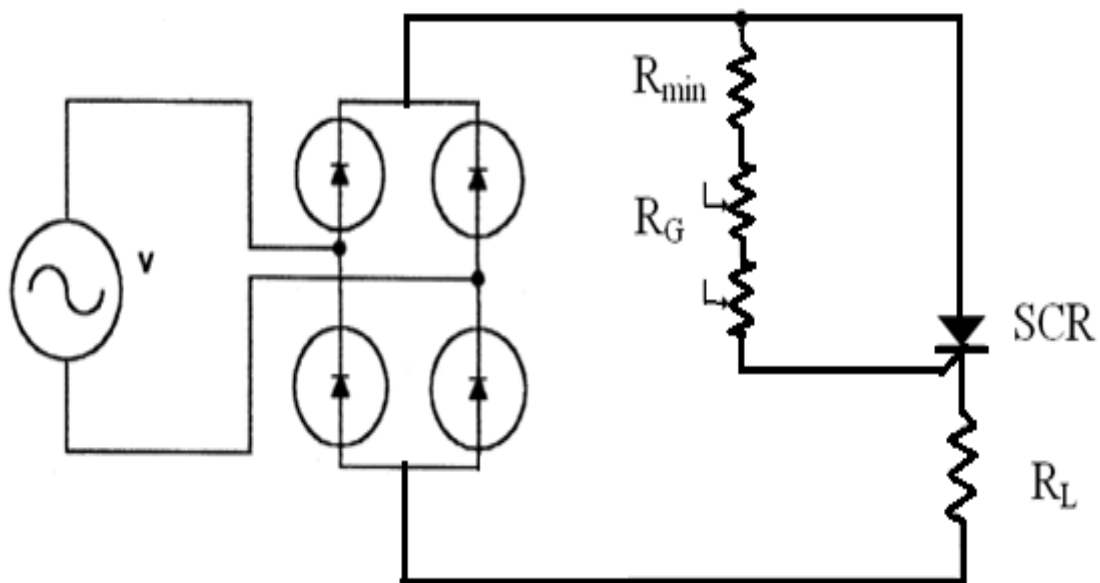


Figure 2

Task:

Vary the firing angle (0 to 90 degree) by changing the gate resistance and observe load voltage and voltage across thyristor on the oscilloscope.

Design Procedure

The design equations are the same as used above.

- Load Resistance = _____ ohms
- R_{min} = _____ ohms
- Gate resistance = _____ ohms
- Variable Resistance = _____ ohms

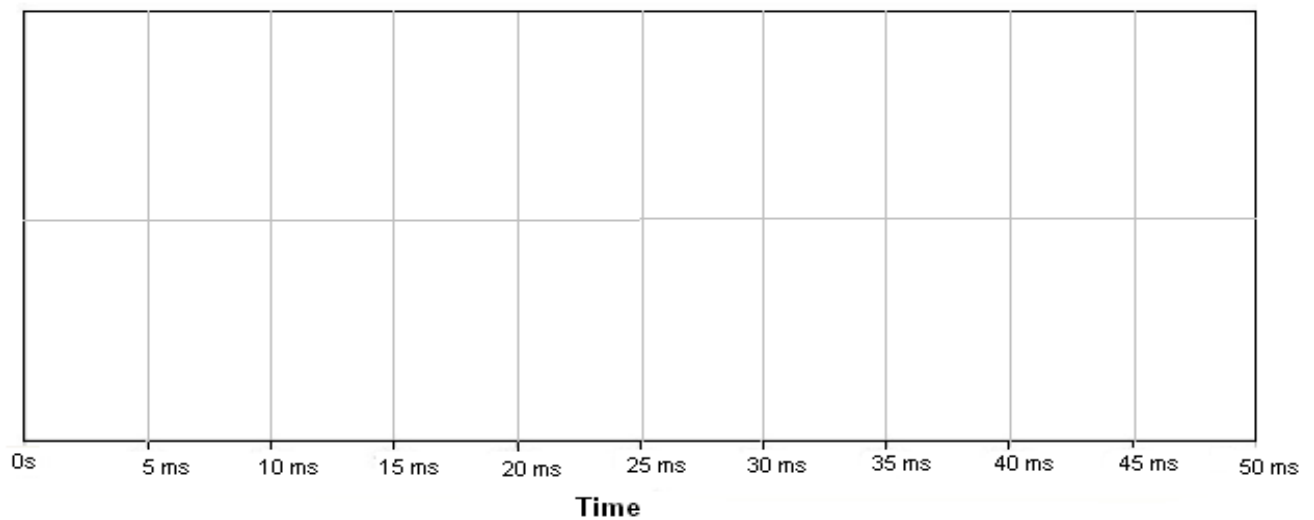
Procedure:

1. Repeat the steps mentioned in section 1.

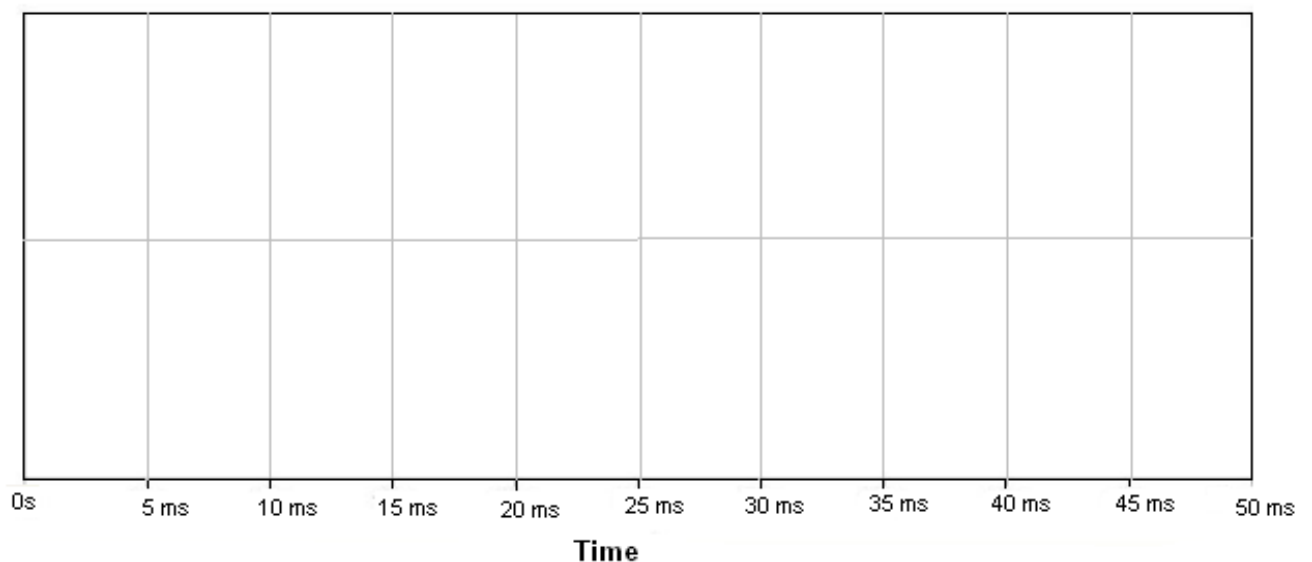
Observation Table 2:

No.	Firing Angle α	V_o (dc) (Measured from Spice Simulations)	V_o (dc) (Theoretically using calculations)
1.			
2.			

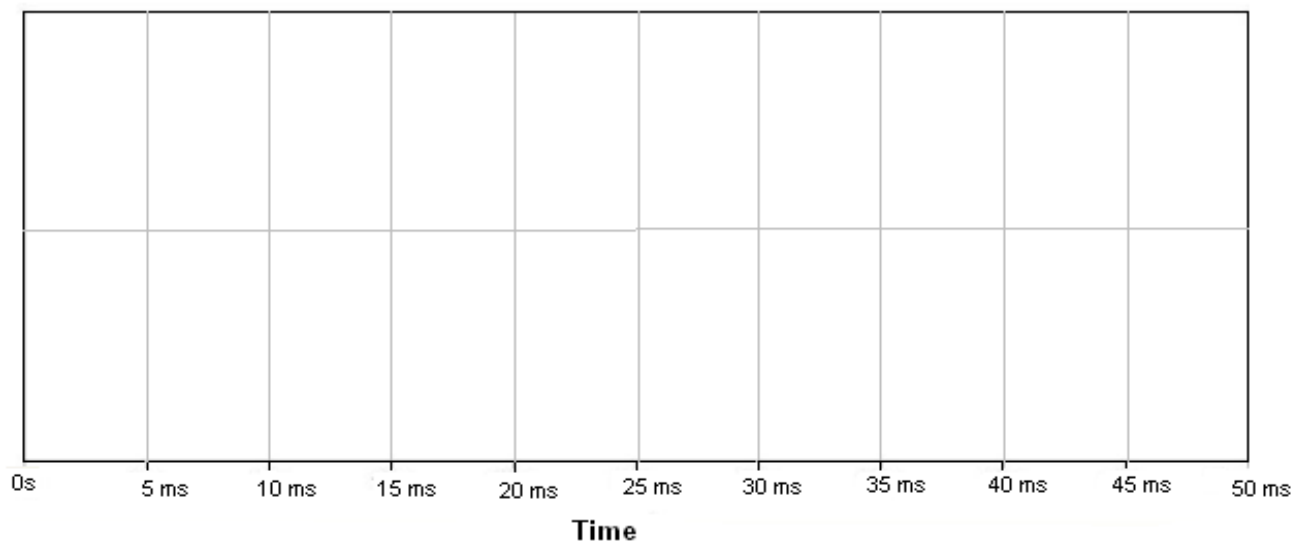
1- Sketch voltage waveform across load resistance for 45° firing angle.



2- Sketch voltage waveform across thyristor (V_{AK}) for 45° firing angle.



3- Sketch input current through AC power supply for 45° firing angle.



Show your calculations of output voltage with both the firing angles:

Please identify if there is any need for free-wheeling diode in this application?

Section3: Three-Phase Half Wave Semi-Controlled Rectifier using R-Triggering

Repeat the above performed simulations for a three-phase half wave rectifier preceded by the same SCR. Record your observations below: